Selection and application of peptide-binding peptides

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Peptide-binding ligands would be useful for directing reagents to particular epitopes in a protein, the detection of peptide hormones, and many other applications. Here we show that peptides of modest size isolated from a library using a simple genetic assay can act as specific receptors for other peptides. The equilibrium dissociation constants of these peptide–peptide complexes are higher than those of typical monoclonal antibody–epitope complexes. Nonetheless, as shown here, these peptide-binding peptides can be used to detect or purify proteins containing the partner peptide.

Keywords: peptides, epitope binding, two-hybrid assay, lambda repressor

Antibodies are extremely useful molecules in medical diagnostics, biotechnology, and biomedical research because of their ability to recognize epitopes with high affinity and specificity. However, they are tedious and expensive to generate, and it is difficult to produce them in very large quantities. As globular proteins, they are relatively fragile molecules and are unsuitable for certain field applications. Therefore, an important goal is to develop nonmacromolecular species that retain the favorable molecular recognition characteristics of antibodies but that can be identified quickly and easily and synthesized in large amounts. Natural peptide-binding proteins or protein domains have been mutagenized to derive species with novel binding specificities[1], but like antibodies, these are globular macromolecules. Although there have been important advances in the development of synthetic receptors in the last few years[2–10], the field remains in its infancy.

Here, we report a simple genetic selection protocol by which peptide libraries can be screened for molecules that bind a particular target peptide sequence. We demonstrate that peptides isolated from these libraries can recognize target sequences in vitro with excellent specificity. Furthermore, these peptide–peptide interactions are more stable than might have been predicted, with equilibrium dissociation constants (K_ds) in the low to medium micromolar range. These affinities are modest compared with those of monoclonal antibody–peptide epitope interactions; however, we nonetheless demonstrate that they are sufficient to support certain western blot–like experiments and affinity purification of proteins.

Results and discussion

A λ repressor-reconstitution assay for the discovery of heteromeric peptide complexes. To screen a peptide library for molecules that bind selectively to a target peptide, we adapted the selection scheme of Hu et al.[11] for the analysis of leucine zipper interactions. Binding of lambda repressor dimers to cognate operator sites is reduced dramatically when the C-terminal dimerization domain is deleted. However, high-affinity repressor–operator binding can be reconstituted by fusion of the DNA-binding domain (DBD) to a number of heterologous proteins or protein fragments that self-associate and functionally replace the native dimerization domain[12–15]. To adapt this assay for our purposes (Fig. 1A), we constructed a plasmid that encoded the DBD fused to a target peptide (an epitope of 13–14 amino acids). A second plasmid construct encoded a fusion of the DBD with a library of DNA fragments. Interaction of the library-encoded peptide (LEP) with the target peptide should reconstitute repressor–operator binding (Fig. 1B). *Escherichia coli* cells carrying a functional repressor are resistant to phage lambda infection, because the repressor blocks the high-level transcription of genes essential for lytic growth[16]. Therefore, cells expressing LEPs that bind the target peptide should grow in the presence of saturating amounts of phage lambda, providing a convenient selection scheme for the desired heterodimers.

A peptide library was prepared by ligating fragments from a *Sau*3A1 digest of chicken genomic DNA into the unique *Bam*HI site downstream of the DNA encoding the DBD repressor (Fig. 1A). It was estimated that the library contained approximately six million clones with genomic inserts. As the *Sau*3A1 restriction fragments are of different sizes, the resultant fusion proteins will contain peptides of different lengths, with the average being determined mostly by the random appearance of a stop codon. Most of the library (>90%) will not encode peptides native to the chicken proteome because half of the restriction fragments are fused in the reverse orientation, two-thirds are out of frame, and much of the DNA is noncoding.

Selection of peptide-binding peptides. The first target employed, NEAYVHDGPVRSLN, is almost identical to a region of the immature form of the interleukin-1β hormone that includes the site at which the prohormone is cut by interleukin converting enzyme (ICE)[17]. An oligonucleotide encoding this ICE cleavage site epitope (ICS) was inserted into the target vector (Fig. 1A), which was then cotransformed into *E. coli* with the chicken DNA-derived library. After a recovery period, the cells were challenged with lambda phage (multiplicity of infection [MOI] = 100) deleted for the wild-type repressor. The 20 resistant colonies were isolated for further analysis. Library-encoded peptides that can self-associate are also expected to give rise to resistant colonies[18]. Subsequent screening for phage resistance in the presence and absence of the target plasmid confirmed at least nine of the colonies as true positives (data not shown).

DNA sequencing identified four different LEPs (two copies of LEPA, one of LEPB, four of LEPC and two of LEPD). Only a partial sequence of the LEPA-containing fusion was obtained (Fig. 2 and 3), because the "peptide" proved to be approximately 23 kDa. Although this is probably a fragment of a chicken protein, as it is unlikely that a stop codon would not occur randomly in such a long sequence, a BLAST search of the genomic databases did not identify a match. One of the other peptides (LEPC) was a mere tripeptide (Pro-Cys-Pro), and the other two positive LEPs identified were 15 and 35 residues (Fig. 2).

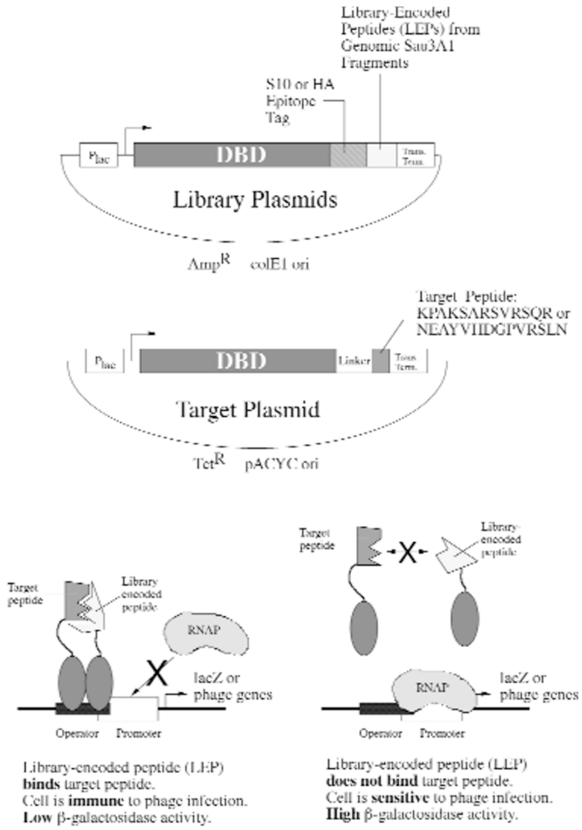

Figure 1. (A) Schematic diagram of the plasmids employed in this study. (B) Overview of the repressor reconstitution assay for screening libraries for peptide-binding peptides. If the LEP binds the target peptide, high-affinity operator binding activity is reconstituted, rendering cells containing this heterodimer resistant to phage infection. If the peptides do not interact, the repressor DNA-binding domains do not have sufficient affinity to bind operator DNA stably, and the cells are sensitive to phage infection.

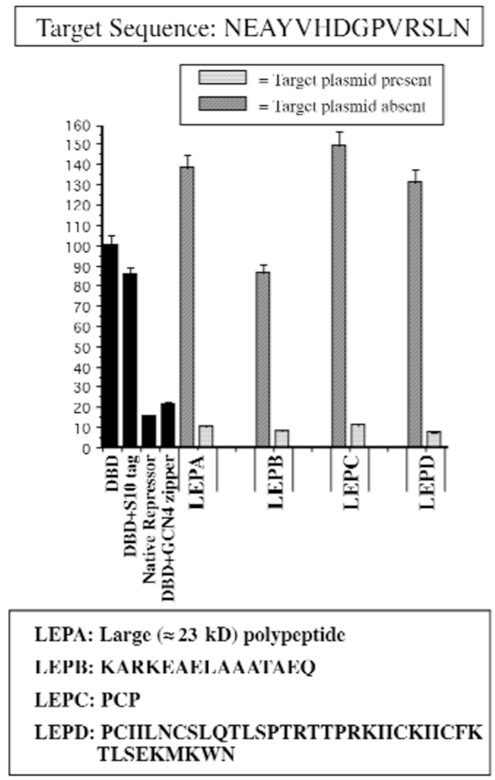

Figure 2. Results of a selection for peptides that bind the target peptide NEAYVHDGPVRSLN, called ICS (see text for details). The cells contain an integrated *lacZ* under the control of a strong promoter with a single overlapping operator site. Cells transformed with plasmids encoding both the LEP and the target peptide had very low β-galactosidase levels, comparable to cells that express native repressor or the repressor DBD fused to the leucine zipper of the GCN4 protein. High levels of *lacZ* expression were observed in the absence of the target plasmid.

To estimate the strength of the putative peptide–peptide interactions, reporter gene experiments were carried out using the integrated repressor-regulated *lacZ* gene carried by this strain[11]. The results (Fig. 2) were consistent with stable binding of the selected peptides to the ICS target peptide in vivo: High levels of transcription were observed in the absence of the target plasmid, but strong repression was detected in the presence of both the target and library-selected plasmids. The peptide interactions appeared to support repressor DBD dimerization about as well as a GCN4 leucine zipper dimer[19] or the native repressor dimerization domain.

In vitro binding studies. In vitro titration experiments were carried out to determine the K_d of one of these peptide–peptide complexes. We prepared a fusion protein comprising Glutathione-S-Transferase (GST) linked to the ICS peptide by a short linker. The fusion was bound to glutathione-agarose beads and incubated with a crude *E. coli* extract from cells expressing the repressor DBD–S10–LEPB fusion protein (Fig. 1A and 2). Determination of the amount of DBD-S20-LEP retained on the beads at different concentrations of GST–ICS indicated that the K_d was approximately 2.1 µM (Fig. 3). Although this value is much higher than typical K_ds of monoclonal antibody-peptide epitope complexes (10^{-8}–10^{-9} M), we note that the interacting ICS and LEPB peptides contain only 14 and 15 residues, respectively. Similar experiments were carried out using LEPA and LEPD fusions. The estimated K_d for the LEPA–ICS complex was approximately 2 µM, although the binding curve showed some nonlinearity, possibly indicating that this is not a 1:1 complex. Library-encoded peptide D bound the ICS more weakly, with a K_D of approximately 30–50 µM (data not shown). Library-encoded peptide C was not tested, because binding could not be detected in a qualitative assay (see below). No binding to GST was observed in the absence of an LEP.

Peptide–peptide interactions can support immunoblot analyses. One of the prime motivations for identifying peptide-binding peptides is to employ them as antibody substitutes. One of the most useful applications of antibodies is the selective detection of proteins in complex mixtures. Therefore, we determined whether ICS–LEP complexes could be used for an immunoblot analysis. Crude extracts were made from cells expressing the repressor DBD–S10–LEP(A–D) fusion proteins or the parent repressor DBD–S10 construct lacking a LEP. These were subjected to SDS–PAGE and transferred onto nitrocellulose membranes. One membrane was probed with anti-S10 antibody to identify the epitope-tagged repressor DBD fusion proteins (Fig. 4). The other membrane was probed with GST–ICS followed by anti-GST antibody to identify bands to which GST–ICS was bound. Signals corresponding to the repressor DBD fused to LEPA and LEPB were easily detected by GST–ICS (Fig. 4). Weaker binding to LEPD was detected, consistent with the fact that the latter binds the target peptide about 10-fold more weakly than LEPA and LEPB. Binding to LEPC was not detected. There was no detectable interaction of GST–ICS with extracts from cells lacking a chicken DNA-encoded peptide. Binding was highly specific, with little detectable association of GST–ICS with other proteins in the extract. Thus, these peptides mimic aspects of antibody–epitope interactions and could be used to detect proteins in crude extracts. However, the modest K_ds do enforce some limitations. We used buffers of lower ionic strength than are typically employed, so as to minimize dissociation of the peptide–peptide complexes during the many washing steps. In addition, experiments using extracts containing lower levels of the repres-

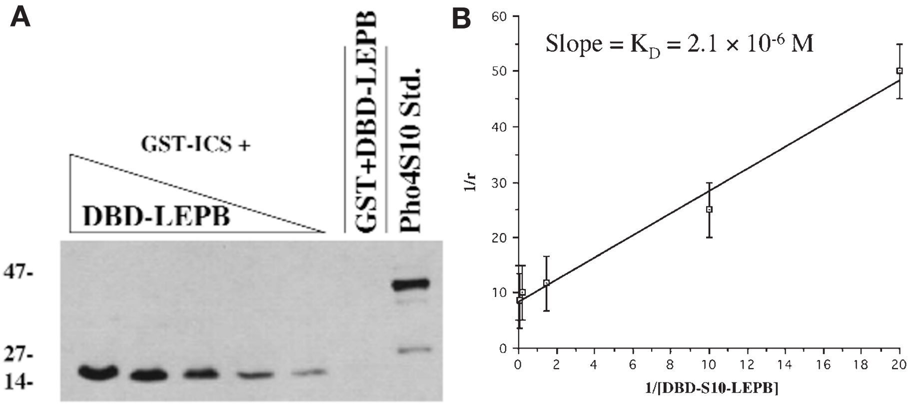

Figure 3. Determination of the K_d of the LEPB-ICS complex. (A) Western blot analysis of proteins bound to GST-ICS beads following incubation with increasing volumes of an extract containing a known amount of the DBD–S10–LEPB fusion protein. GST lacking the ICS peptide did not retain DBD–S10–LEPB (second to last lane). (B) Double reciprocal plot from which a K_d of approximately 2.1 μM was obtained.

sor DBD–peptide fusion demonstrated that the LEPs are not as sensitive as the monoclonal antibody in this assay (data not shown).

Affinity purification of a tagged protein based on peptide–peptide interactions. A second selection was performed using a different target, a 13-residue sequence from an internal region of the immature human insulin-like growth factor I (IGFI) prohormone (KPAKSARSVRAQR). This experiment employed a library constructed from approximately 50,000 yeast genomic *Sau*3A1 restriction fragments, and yielded a single positive that was not a result of self-association of the LEP[18]. The amino acid sequence inferred from the DNA sequence was: THTTSQTTLRDPDVYAGARWVTWRVGA. The underlined residues are encoded by yeast DNA, and the remaining residues are vector-encoded. As a result of a cloning artifact that caused a frameshift, the vector-encoded residues do not express the haemagglutinin (HA) epitope tag that had been designed into the library vector. As these amino acids are not present in the repressor DBD-target peptide construct, they could be involved in heterodimerization and are therefore considered to be part of LEP27, although they may or may not be important for binding.

Using purified fusion proteins MBP– THTTSQTTLRDPDVYAGARWVTWRVGA and GST–WKPAKSARSVRAQR, the K_d of the 27-mer–13-mer complex was found to be approximately 20 μM (data not shown). As mentioned previously, a potentially useful application of these complexes, in addition to immunoblot analyses, is affinity purification of proteins. To test this, we adsorbed the purified MBP–LEP27 chimera to an amylose column. To this column, we applied a crude extract prepared from *E. coli* cells that expressed the GST–13-mer target peptide chimera. The column was washed thoroughly with buffer, and the MBP–LEP27 fusion and any associated proteins were eluted with maltose. The only proteins detected in the maltose-eluted fraction were the GST-target peptide and MBP–LEP27 (Fig. 5). Control experiments demonstrated that this association required the presence of both the target IGFI prohormone-derived sequence and LEP27. The GST-target peptide was not retained from the *E. coli* extract by either MBP alone or a fusion containing a 60-residue peptide unrelated to LEP27. GST) lacking the target peptide fusion did not bind to an MBP–LEP27 column (Fig. 5).

These results demonstrate that peptides of modest size can function as selective peptide-binding molecules. Low to medium micromolar K_ds were observed. Although they were much weaker than a strong antibody–epitope contact, they were nonetheless sufficient to allow the peptides to be used as antibody equivalents in some applications. For affinity-purification experiments, the modest K_ds of peptide–peptide complexes may be an advantage. Library-encoded peptide concentrations well in excess of micromolar amounts can be achieved in a column format; therefore, efficient retention of the target protein should be possible. The modest binding affinity should also allow the elution of the protein of interest from the column under mild conditions. For applications in which higher affinities are desirable, it may be possible to mutate these first-generation LEPs to identify variants that bind their target more tightly.

The repressor reconstitution assay[12] we used to isolate peptide-binding peptides is one of many variants of the two-hybrid concept[20–22]. Because of the complications of the "homodimer" background, this assay is not employed as widely as many other two-hybrid assays for probing protein–protein interactions[18]. However, in our hands, neither the standard yeast two-hybrid assay nor phage display

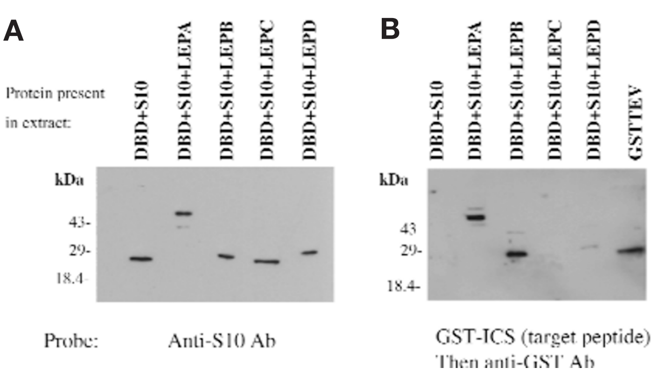

Figure 4. A far western blot reveals direct interactions between the ICS and the LEPs selected from the chicken library. (A) Western blot using the anti-S10 antibody. (B) Far western blot in which the gel was first probed with GST–ICS, then anti-GST antibody. The final lane contains a purified GST fusion protein, which was employed as a control to ensure that the anti-GST antibody bound GST-ICS retained on the membrane.

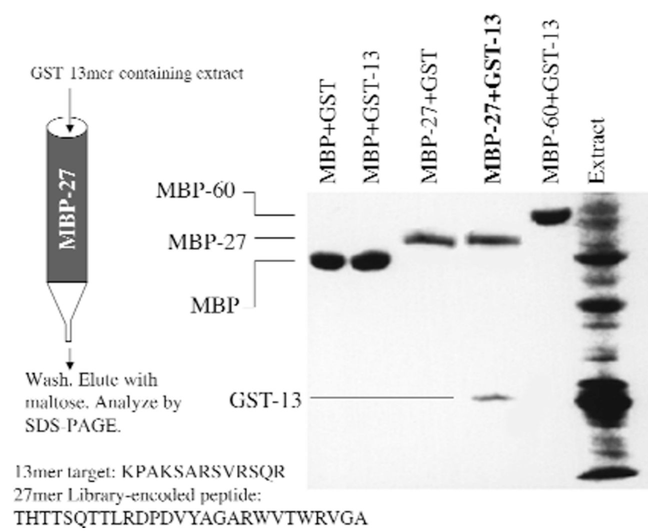

Figure 5. Purification of GST–13-mer by affinity chromatography using immobilized MBP–LEP27. Shown is a Coomassie brilliant blue-stained denaturing polyacrylamide gel. A crude extract made from *E. coli* cells expressing either GST or the GST-target peptide fusion protein was applied to amylose columns saturated with either MBP, MBP-LEP27, or MBP-60a.a. (a fusion containing a 60 residue peptide unrelated to LEP27). Bound proteins were eluted with maltose.

identified peptide-interacting peptides (Z.Z. and T.K., unpublished observations). One might speculate that as the peptide-peptide interaction is templated by the two DBD-operator half-site binding events, this assay may be more sensitive than others for the detection of interactions with K_ds in the micromolar range. Since interactions between peptide functional groups cannot easily be shielded from solvent, as is the case when an epitope occupies an antibody cavity, it is important to employ an assay capable of identifying interactions of modest affinity.

Experimental protocol

Target plasmids. Vectors encoding the repressor DBD fused to the target peptides were made by inserting the appropriate oligonucleotides into pJH391 [**PLEASE CONFIRM**] (ref. 11) digested with *Bam*HI and *Sal*I. The resulting plasmids were further digested with *Eco*RI and *Eco*RV to generate a DNA fragment that was inserted into pACYC184 (Bio-Rad Laboratories, Hercules, CA) digested with *Eco*RI and *Pvu*II.

Construction of the library. An S10 epitope tag–encoding oligonucleotide was inserted into pJH391, providing pJH391s. The library was made by inserting chicken genomic *Sau*3A1 restriction fragments into *Bam*H1-digested, phosphatased pJH391S. Subsequent characterization suggested that the library contained between five and six million clones with inserts. Details of the library construction and characterization are available upon request.

Selections. *Escherichia coli* cells (strain JH372) harboring pICS were transformed with the repressor DBD–S10 epitope–peptide fusion protein-encoding library. The transformants were challenged with lambda phage K54 and K54H80 (MOI of 100; kindly provided by Professor James Hu, Texas A&M University, College Station, TX), lacking a functional repressor protein. To segregate survivors that contain library plasmids encoding self-associating peptides from those with target-binding peptides, the survivors were cured of the target plasmid and tested for phage resistance. As a complementary test, the β-galactosidase activity[23] of each the survivors was assayed. JH372 has an integrated *lacZ* gene with a promoter containing a single high-affinity lambda operator[11]. Gene expression was measured both in the presence and absence of the target plasmid. Full details of the selection protocol are available upon request.

Fusion proteins. Oligonucleotides encoding the desired peptide were inserted into pGEXcs (ref. 24) digested with *Nco*I and *Bam*HI (GST fusions) or into pMALc2 vector (New England Biolabs, Beverly, MA) cut with *Nco*I and *Bam*HI (MBP fusions). The GST fusion proteins were expressed in BL21 *E. coli* and purified by glutathione agarose affinity chromatography, or amylose affinity chromatography.

Far western blot. Cells expressing the repressor DBD–S10–LEP fusion proteins were induced with isopropylthiogalactoside (0.5 mM). One milliliter of an overnight culture was centrifuged, the pellet was boiled in gel loading dye, and an aliquot was electrophoresed through a 20% SDS polyacrylamide (tricine) gel. The samples were then transferred to a nitrocellulose membrane and soaked in 8 M urea in Far Western (FW) buffer (20 mM TRIS HCl, pH 7.5, 0.1 mM EDTA, 5% glycerol, 0.04% NP-40, and 40 mM NaCl) for 30 min. The membrane was then washed successively with FW buffers containing 4 M, 1 M, 0.5 M, 0.1 M, and 0 M urea, respectively, at 4°C. The membrane was blocked with 5% dry milk and washed three times with FW buffer. The membrane was incubated with 1 ml of 6 mg/ml GST–ICS overnight at 4°C, then washed with FW buffer for 10 min at 4°C. The membrane was then probed with mouse anti-GST IgG (Santa Cruz Biotech, Santa Cruz, CA) for 2 h at 4°C, then washed at the same temperature with FW buffer. Finally, the membrane was incubated with goat antimouse IgG–AP conjugate (Bio-Rad) for 2 h at 4°C, washed with FW buffer, incubated with SuperSignal West Dura (Pierce Chemical Company, Rockford, IL), and subjected to autoradiography.

Estimation of dissociation constants. A crude extract was prepared from cells that expressed DBD–S10–LEPB fusion protein. The amount of tagged protein in the extract was determined by quantitative western blot analysis using purified S10Pho4p as a standard (provided by L. Sun, University of Texas Southwestern Medical Center, Dallas, TX). A series of pull-down assays using GST–LEPB was then carried out to determine the fraction of tagged protein bound at various LEPB concentrations. The data were plotted as described by Freifelder[25], using the equation: $1/r = K_d/(\text{DBD–S10–LEPB}) + 1$, where $r = (\text{DBD–S10–LEPB})_{bound}/(\text{GST–ICS})_{total}$. The slope of the line obtained in a double reciprocal plot provides the K_d. A similar protocol was employed to measure all other K_ds reported here.

Affinity chromatography using MBP–27-mer fusion protein. A 0.5 ml column of amylose beads saturated with the MBP–LEP27 fusion protein was prepared and equilibrated with column buffer. Two hundred milliliters of concentrated extract prepared from a 6 L culture of cells expressing the GST-target peptide fusion protein was applied to the column at room temperature. The flow rate was about 2 ml/h. Afterward, the column was washed with column buffer containing 150 mM NaCl, then 300 mM NaCl. The MBP–LEP11 fusion protein and any proteins bound to it were then eluted with 10 mM maltose in column buffer. After dialysis, the sample was concentrated, subjected to 10% SDS–PAGE and stained with Coomassie brilliant blue.

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